# Twin Layer: Operations & User Guide

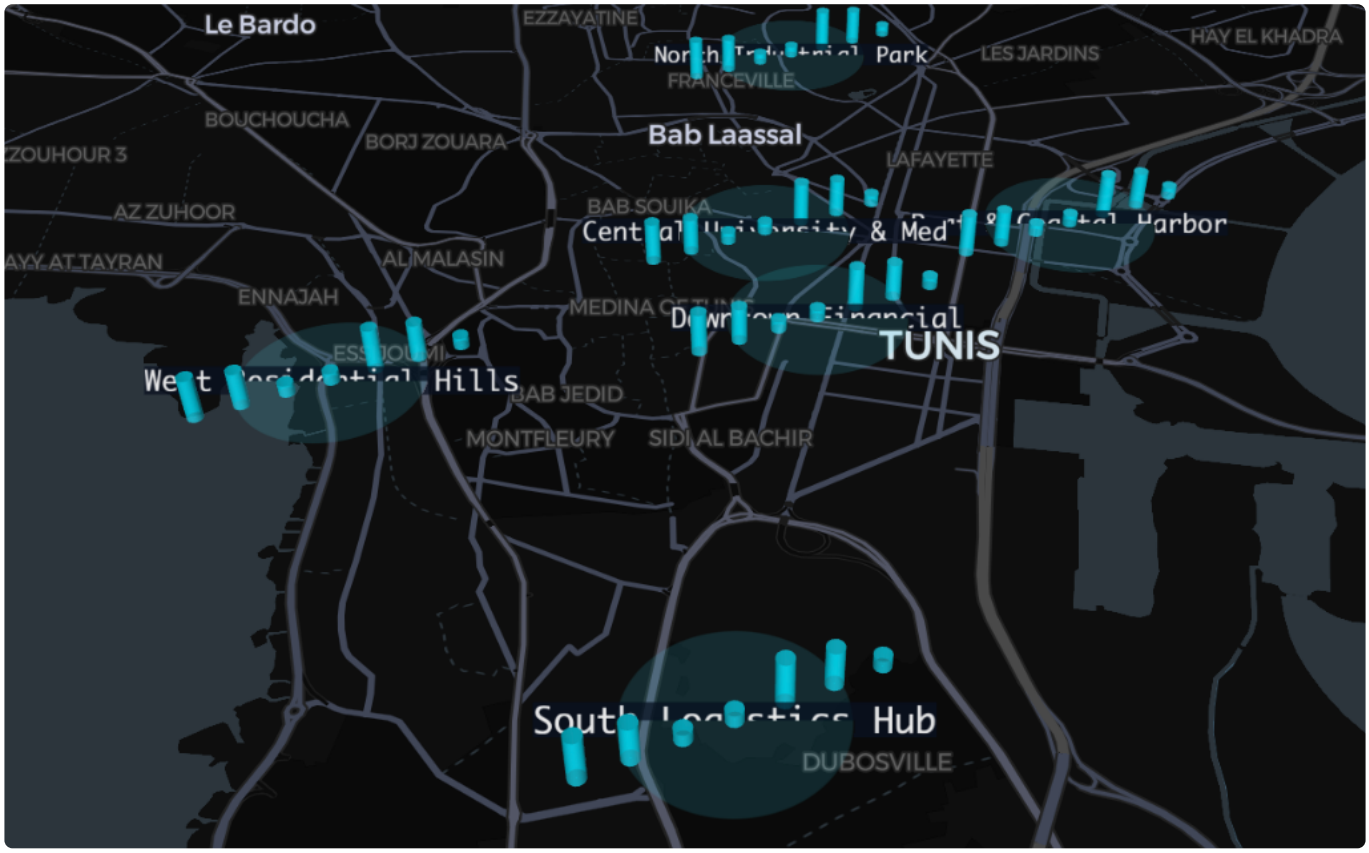
This guide is an illustrated walkthrough of the Twin Layer Adversarial Console. It explains the visual interface, details how each sector and sensor functions, and provides a 5-minute hands-on tutorial for testing all 6 adversarial capabilities.

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## 1. Interface Tour & Visual System

The Adversarial Console runs in dark tactical mode ( `#060913` background with navy glassmorphic panels). The interface uses three status colors:
- **Cyan ( `#00e5ff` ):** Nominal telemetry and healthy sensor pillars.
- **Amber ( `#ffb300` ):** Minor variance, unverified key, or degraded connectivity.
- **Crimson ( `#ff1744` ):** Active adversarial attack, signature failure, or physical state divergence.

### A. 3D Tactical Map (Greater Tunis)



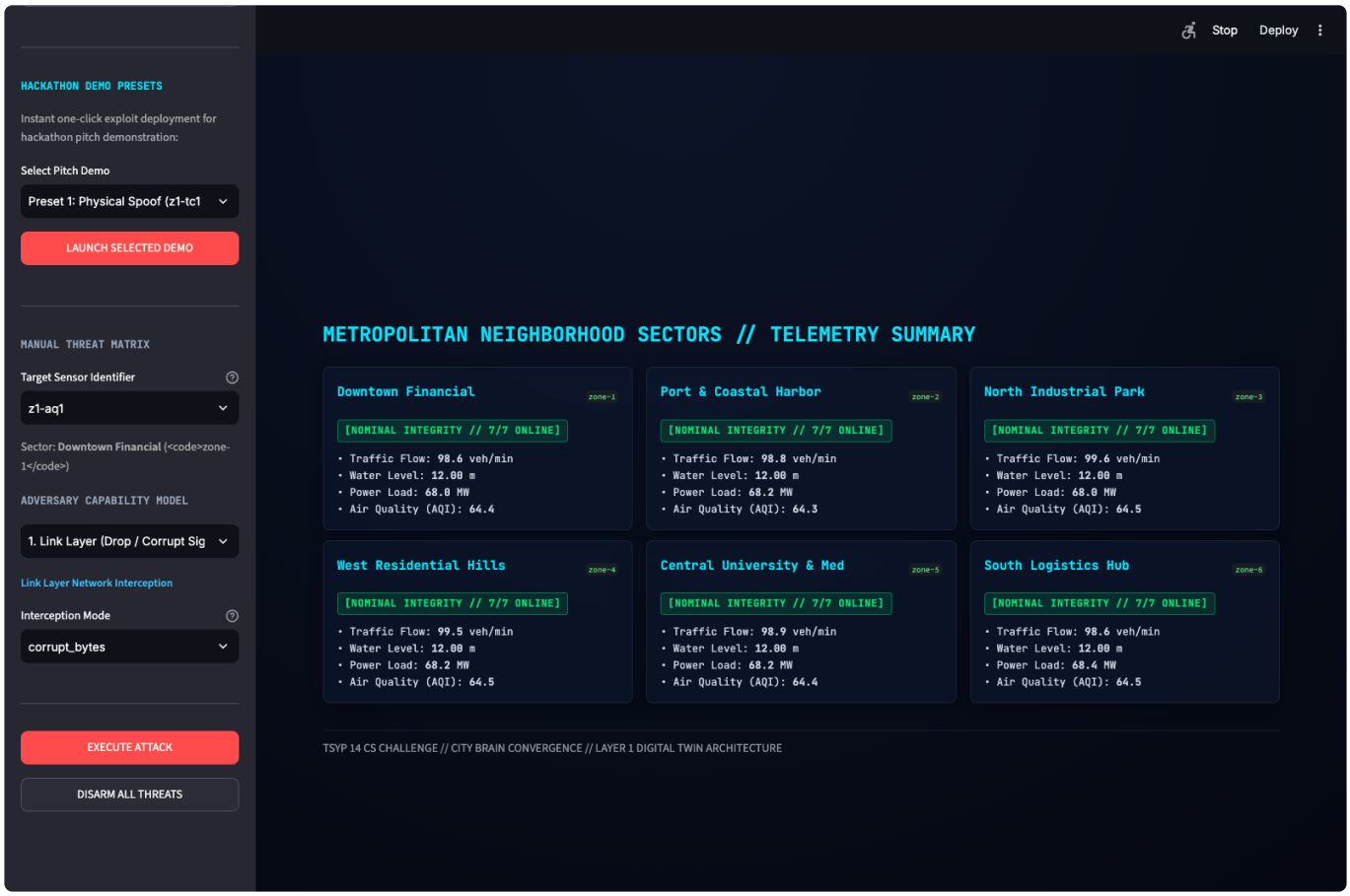

*Greater Tunis 3D Tactical Map*

The map displays all 6 municipal sectors across Greater Tunis, with fixed centroids on dry land:
1. **Downtown Financial ( `zone-1` ):** Commercial center along Avenue Habib Bourguiba.
2. **Port & Coastal Harbor ( `zone-2` ):** Radès and La Goulette harbor docks and shipping canal.
3. **North Industrial Park ( `zone-3` ):** Charguia and Ariana Sud industrial hub.
4. **West Residential Hills ( `zone-4` ):** Residential terrain across Bardo and Manouba hills (positioned north of Sebkhet Sijoumi lake).
5. **Central University & Med ( `zone-5` ):** El Manar campus and Rabta medical district.
6. **South Logistics Hub ( `zone-6` ):** Ben Arous freight rail terminal and supply warehouses.

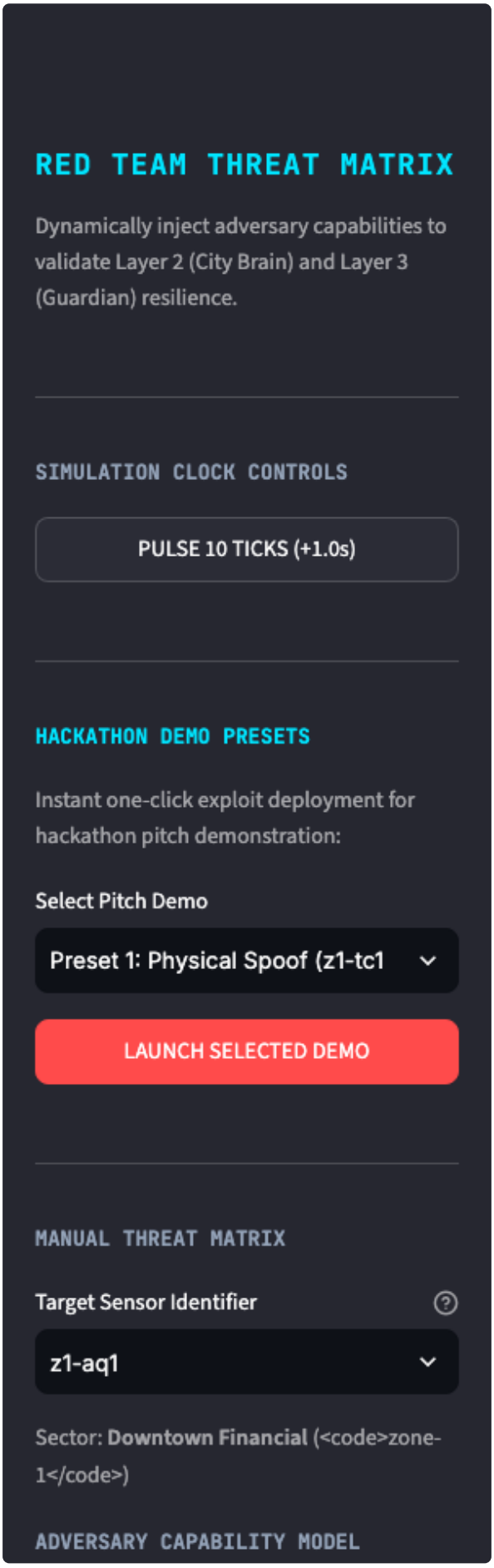
Each sensor is rendered as a 3D hexagonal column whose height scales with its current metric. When a sensor is nominal, its pillar and sector halo glow cyan. When an attack is injected, the target pillar elevates and turns crimson, surrounded by a red warning beacon.

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## B. Neighborhood Operational Cards & Red Team Sidebar




*Metropolitan Neighborhood Sectors and Telemetry Summary*




*Red Team Threat Matrix Controls*

- **Neighborhood Cards (Main View):** Each card summarizes the 7 sensors operating within that zone:
- **Traffic Cameras ( `tc1` , `tc2` ):** Vehicle passage rates (0 - 200 veh/min).
- **Water Level Floats ( `wl1` , `wl2` ):** Drainage channel depth (0 - 12 meters).
- **Power Load Meters ( `pl1` ):** Electrical grid draw (5 - 120 MW).
- **Air Quality Stations ( `aq1` ):** Particulate index (10 - 350 AQI).
- **Emergency Dispatch ( `ec1` ):** Active incoming emergency calls (0 - 30 call/min).
- **Red Team Controls (Sidebar):** Contains attack configuration controls, one-click demo presets, simulation clock pulse controls, and the active threat manifest showing all currently running exploits.

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### C. Clean Vector Cartography



*Clean Vector Cartography*

The basemap uses clean dark vector tiles ( `CARTO Dark Matter` ) without cluttered text labels, allowing clean zooming into street-level intersections without broken fonts or overlapping water bodies.

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## 2. Hands-On Tutorial: Testing the System in 5 Minutes

Follow these steps to test every capability in real time.

### Step 1: Open the Console & Verify Nominal Baseline

1. Open your browser and navigate to `http://localhost:8501` .
2. Inspect the top Scoreboard:

3. **Total Readings:** Actively incrementing.
4. **Active Incidents / Tampered:** 0.
5. **Active Red Team Attacks:** 0 ( `Zero Threats` ).
6. **System Status:** `[SYSTEM STATUS: AUTONOMOUS] // LOCAL SIMULATION RUNTIME ACTIVE` .
7. Verify that all 6 neighborhood cards show green `[NOMINAL INTEGRITY // 7/7 ONLINE]` badges.

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## Step 2: Inject a Physical Sensor Spoof ( `device_physical` )

In this attack, the physical sensor reading is falsified on the wire, but the ground-truth database records the true physical reality.

1. In the sidebar under **"MANUAL THREAT MATRIX"**:
2. Set **Target Sensor Identifier** to `z1-tc1` (Downtown Traffic Camera 1).
3. Set **Adversary Capability Model** to `4. Physical Layer (Sensor Spoofing)` .
4. Enter **Spoofed Telemetry Value**: `185.0` (simulating extreme artificial traffic gridlock).
5. Click **"EXECUTE ATTACK"**.

**What happens immediately:**
- The Downtown Financial card flips to red: `[CRITICAL THREAT ACTIVE]` .
- On the 3D map, the `z1-tc1` pillar elevates and pulses crimson.
- The sidebar **Active Threat Manifest** lists `[THREAT ACTIVE] device_physical (z1-tc1)` .

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## Step 3: Inspect Ground Truth vs Broadcast Divergence

1. Click **Tab 2: "Live Telemetry & Divergence Curves"**.
2. Select Device `z1-tc1` from the dropdown.
3. Observe the Altair time-series chart:
4. **Solid Emerald Line:** Real physical reality (e.g., ~45 veh/min).
5. **Dotted Crimson Line:** Spoofed broadcast value (185 veh/min).
6. **Divergence Metric:** Shows an absolute discrepancy of ~140 veh/min.
7. This divergence proves the physical isolation rule: the external world receives the poisoned data, but the Twin's internal SQLite ledger preserves true ground reality.

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## Step 4: Test the Other 5 Attack Capabilities

You can test each capability individually using the sidebar controls:

| Capability | Target Device | Test Configuration | Observable Result |
|---|---|---|---|
| `device_key` (Compromised Key) | `z3-pl1` (Power Load) | Replaces device key with an attacker key | Outbound signature remains valid, but Guardian detects revoked/unauthorized public key on MCP. |
| `link` (Network Tampering) | `z2-wl1` (Water Float) | Corrupts raw JSON bytes or drops frames | Guardian signature validation immediately fails ( `Ed25519 verification failed` ); payload corrupted. |
| `rogue` (Unregistered Transmitter) | `zone-4` (Residential) | Transmits unauthorized packet `rg-z41` | Packet appears on wire; Guardian MCP tool `list_devices` flags sensor as unregistered intruder. |

| Capability | Target Device | Test Configuration | Observable Result |
|---|---|---|---|
| `sensor_fault` (Hardware Drift) | `z5-aq1` (Air Quality) | Injects additive drift (`+45.0 AQI`) | Values drift upward without triggering hard boundaries; caught by multi-sensor neighborhood correlation. |
| `actuator` (Command Disobedience) | `z1-wl2` (Flood Valve) | Sets actuation state to `stuck_closed` | Device returns `"status": "ok"` to City Brain MCP call, but physically refuses to open drainage valve. |

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## Step 5: Use One-Click Hackathon Pitch Presets

During live presentations, you do not need to manually configure parameters. The sidebar includes **6 Pitch Presets** pre-calibrated for demonstration:

1. Click **"DEMO PRESET 1: PHYSICAL SPOOF (DOWNTOWN)"**.
2. Spikes traffic camera `z1-tc1` to 195 veh/min.
3. Demonstrates immediate detection on 3D map.
4. Click **"DEMO PRESET 2: FORGED KEY (INDUSTRIAL)"**.
5. Emulates key theft on power substation `z3-pl1`.
6. Demonstrates zero-trust key revocation.
7. Click **"DEMO PRESET 3: LINK CORRUPTION (HARBOR)"**.
8. Injects bitflips into harbor water level `z2-wl1`.
9. Demonstrates signature verification failure.

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## Step 6: Disarm Threats & Confirm Baseline Recovery

1. In the sidebar, click the button **"DISARM ALL THREATS"**.
2. **Verification:**
3. Active attacks count immediately resets to `0` (`Zero Threats`).
4. All 6 neighborhood cards turn back to `[NOMINAL INTEGRITY // 7/7 ONLINE]`.
5. All 3D pillars return to uniform cyan baseline.
6. The Active Threat Manifest clears completely.

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## Step 7: Forensic Audit Ledger

1. Click **Tab 3: "Raw Ledger & Forensic Audit"**.
2. The table shows the immutable record of all injected attacks stored in `attacks_injected`:
3. Exact simulation timestamp (`start_sim_time` and `end_sim_time`).
4. Attack type, target device, and attack parameters.
5. Active status (`1` when running, `0` once disarmed).
6. This audit trail gives judges and security evaluators proof of attack lifecycles and response timestamps.